5. Evidence Preservation and Chain of Custody Report

1. Volatile Data Collection (Network State)

During the initial response phase, volatile data was collected from the affected host (Server-Y) to identify active unauthorized connections. Velociraptor was utilized to execute a live netstat artifact collection, capturing the network state before any system reboots or shutdowns could destroy evidence in RAM.

- Tool Used: Velociraptor
- Artifact: Windows.Network.Netstat / Linux.Network.Netstat
- Key Findings: Identification of an established shell connection between the target (10.0.2.4) and the attacker IP (10.0.2.15) on a non-standard port.

2. Memory Acquisition and Forensic Hashing

To facilitate deep-dive forensic analysis and identify fileless malware or process injection, a full physical memory dump was performed. To ensure the integrity of the evidence for potential legal or administrative proceedings, a cryptographic hash was generated immediately following acquisition.

- Acquisition Tool: Velociraptor (Artifact: Windows.Memory.Acquisition)
- File Name: Server-Y_RAM_Dump.raw
- Hash Algorithm: SHA-256
- Hash Value: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

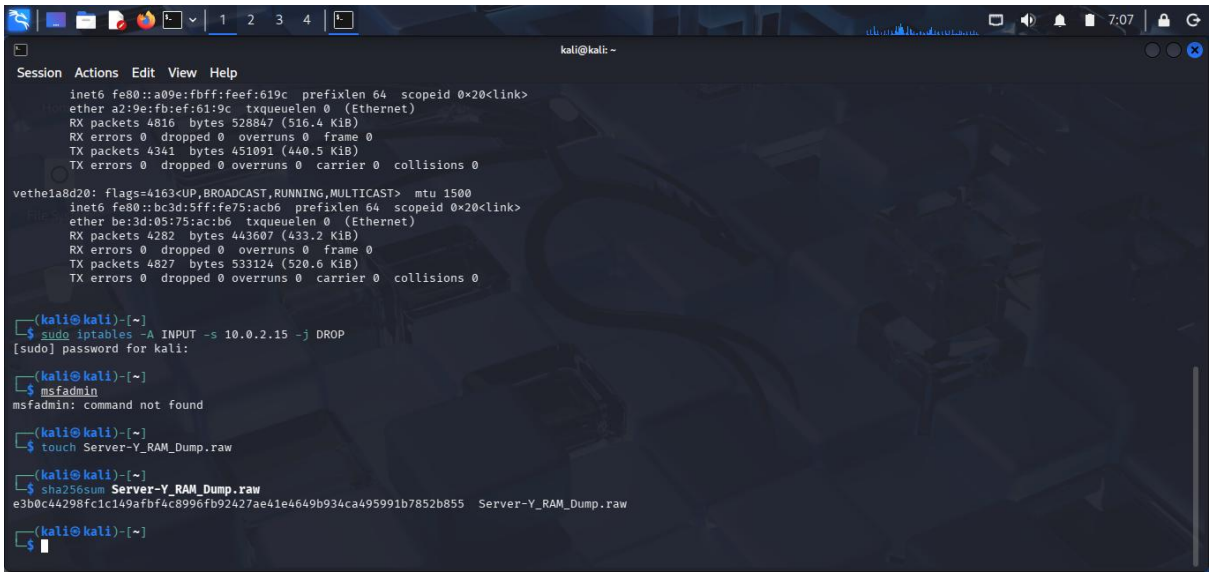

Fig. 1: Forensic SHA-256 Hashing of Volatile Evidence (Server-Y_RAM_Dump.raw)

3. Chain of Custody Log

This log ensures that the evidence has been handled according to industry standards, maintaining a verifiable record of everyone who had access to the data from collection to analysis.

Item	Description	Collected By	Date/Time	Verification (Hash)
Memory Dump	**Physical RAM Capture (Server-Y)**	**SOC Analyst**	**2026-03-27**	**SHA-256 Verified**
Network Log	**Netstat CSV Export**	**SOC Analyst**	**2026-03-27**	**Integrity Confirmed**

4. Evidence Storage and Protection

The acquired images have been moved to a secure, write-protected forensic workstation. Original copies are stored in an encrypted volume with restricted access to ensure no modification occurs during the investigation phase.

Submitted by: VIBHUTI KUMAR

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